When the membership and homogeneous-existence loci coincide, and a proof that they do for ansatz 5 on hydrogen

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A companion note to *A New Method of Solving PDEs*, and a sequel to *Membership versus variety, and the partial-solution strata* and *Recovering the partial-solution strata by membership alone: a GCRD closure construction*. The core algorithm computes the **membership locus** $V_\forall(A) = \{c : P \in [A(c)]\}$; the existential computation on $A \cup \{P\}$ computes the **homogeneous existence locus** $V_{\exists\setminus\Psi}(A) = \{c : 1 \notin [A(c), P] : \Psi^\infty\}$. This note (i) records the exact condition under which the two coincide, (ii) proves coincidence unconditionally for one-dimensional ansätze, and (iii) proves by an explicit greatest-common-right-divisor (GCRD) computation that *ansatz 5 on the hydrogen Schrödinger equation has no genuine partial strata*, so that the two loci coincide there off the bad locus, and the whole gap between them is confined to the ODE-degeneration loci that the differential Thomas decomposition already resolves cell by cell.

1 The exact gap, and its GCRD reading

The relationship between the two loci was established in the two companion notes. Writing the existential projection as membership plus a remainder,

$$V_{\exists\setminus\Psi}(A) \;=\; V_\forall(A) \;\sqcup\; \{\text{partial-solution strata}\}, \tag{1}$$

where a *partial-solution stratum* (companion note I, Definition 1) is a c at which a proper nontrivial subfamily of $\mathrm{Sol}(A(c))$ satisfies P while the family as a whole does not. Hence

$$\boxed{V_\forall(A) = V_{\exists\setminus\Psi}(A) \iff A \text{ has no partial-solution strata.}} \tag{2}$$

For an ansatz whose top element is a *linear* ODE L of order n in the inner variable v, companion note II sharpens this into an order condition. Ritt-reducing P modulo A and splitting the remainder over base power products yields operators $R_1, \ldots, R_s \in K(v)[\partial]$ of order $\le n-1$; with

$$G(c^*) \;=\; \mathrm{GCRD}\big(L(c^*), R_1(c^*), \ldots, R_s(c^*)\big), \qquad k = \operatorname{ord} G(c^*),$$

the trichotomy (note II, Theorem 1) reads: $k = n \iff c^* \in V_\forall$ (the whole family solves P); $k = 0 \iff$ no common solution; and $0 < k < n \iff c^*$ is a partial stratum. Therefore

$$V_\forall(A) = V_{\exists\setminus\Psi}(A) \quad \text{off the bad locus} \quad \iff \quad \operatorname{ord} G(c^*) \in \{0, n\} \;\; \text{for all such } c^*. \tag{3}$$

Coincidence is exactly the statement that the GCRD never has intermediate order.

Remark 1 (hypotheses and their necessity)**.** *Everything past* (2) *rests on two standing hypotheses, and both are load-bearing.*

(H1) P is linear and homogeneous in Ψ. *Linearity is what makes the reduced remainder a* linear *operator:* $r = \sum_\beta m_\beta R_\beta(\Psi)$ *with* $R_\beta \in K(v)[\partial]$*, the object whose GCRD with* L *the whole analysis takes. A* Ψ^2*- or* $|\Psi|^2\Psi$*-term would make* r *nonlinear in the jets, and there would be no operator to divide — the Ore ring, the kernel-intersection identity, the Wronskian argument and the subresultant strata equations all require* r *to be an operator. Homogeneity is what makes the* P*-solving solutions a linear* subspace *of* $\mathrm{Sol}(A(c^*))$*, on which Lemma 1 turns; without it a one-dimensional family could solve* P *at isolated scalings and the dimension argument would fail. Linearity here means linear in* Ψ *and its derivatives; arbitrary coefficients in the independent variables are allowed, so the Coulomb* $-1/r$ *is fine and the Schrödinger equation satisfies (H1). A nonlinear Schrödinger / Gross–Pitaevskii equation does not, and lies outside the theory of this note entirely.*

(H2) the ansatz top element is a linear ODE. *This is what places* L *in the right-Euclidean ring* $K(v)[\partial]$*, where GCRDs exist and are computable; the trichotomy is a statement about* ord GCRD *of linear operators. Nonlinear structure* below *the top is admissible provided it is absorbed into the differential base field: the algebraic-extension elements of ansätze 13 and 16, the roots, and the coefficient-ring extensions 14, 15 leave the top element linear (over* $K(v)(g)$*, say), and GCRD theory is valid over any differential field. Companion note II further relaxes (H2) for a* first-order *nonlinear top, which has an algebraic-extension closure of bounded degree; a* higher-order *nonlinear top is open. The one library ansatz that genuinely violates (H2) is the product family 10 (*$\Psi = F(\xi)\,G(\eta)$*): eliminating* F, G *from the nonlinear relation* $\Psi - F\cdot G$ *gives a nonlinear top for* Ψ*, outside* $K(v)[\partial]$*.*

Both hydrogen and helium are linear and homogeneous in Ψ*, so (H1) holds throughout; every library ansatz except 10 satisfies (H2). The ansatz-5 result of §3 sits squarely in the doubly-linear regime.*

2 One-dimensional ansätze coincide unconditionally

The cleanest sufficient condition is dimensional and needs no GCRD machinery.

Lemma 1 (dimension-one coincidence)**.** *Let* P *be linear and homogeneous in* Ψ*. If, for every* c^* *off the bad locus,* $\dim_{\mathbb{C}} \mathrm{Sol}(A(c^*)) \le 1$*, then* $V_\forall(A) = V_{\exists\setminus\Psi}(A)$ *off the bad locus.*

Proof. For P linear and homogeneous, $\{\Psi \in \mathrm{Sol}(A(c^*)) : P(\Psi) = 0\}$ is a *linear* subspace of $\mathrm{Sol}(A(c^*))$. A space of dimension ≤ 1 has only $\{0\}$ and itself as subspaces, so either every ansatz solution solves P (then $c^* \in V_\forall$) or only the zero solution does (then $c^* \notin V_{\exists\setminus\Psi}$). In both cases c^* lies in $V_\forall$ iff it lies in $V_{\exists\setminus\Psi}$. In the GCRD reading, $n \le 1$ forces $k \in \{0, n\}$, so (3) holds. □

The Schrödinger operator is linear and homogeneous in Ψ, so Lemma 1 applies to every ansatz of the library whose solution space is one-dimensional — the first-order-ODE and pure-exponential families. The two-dimensional (second-order ODE) families are not covered, and, as companion note I shows with $A : \Psi'' - c\Psi$, $P : \Psi'' - 3\Psi' + 2\Psi$, they can carry genuine partial strata even when the PDE order matches the ansatz order. Whether a *particular* second-order ansatz has strata against a *particular* PDE is the residual question; §3 settles it for ansatz 5 on hydrogen.

$\dim \mathrm{Sol}(A(c))$	library ansätze	status
1 (1st-order ODE / pure exp)	8, 9, 1, 1.1, 20, 20.1	coincide (Lemma 1)
1? (log additive mode)	2, 17, 17.1, 18, 19	dimension check pending
2 (2nd-order linear ODE)	5, 5.1, 5.2, 5.3, 3, 6, 7, 13, 14, 15, 16	GCRD analysis (§3)
higher / nested	12	GCRD tower recursion
nonlinear (product $F \cdot G$)	10	outside the linear theory

3 Ansatz 5 on hydrogen has no genuine partial strata

We carry out the GCRD analysis of (3) for the paper's figure "ansatz 5" against the hydrogen Schrödinger equation, and find that the only partial strata sit on the bad locus. The computation is elementary linear algebra over $\mathbb{Q}(c)$; the verification scripts are `ansatz5_gcrd.sage` and `ansatz5_gcrd2.sage`.

3.1 Setup and the reduced remainder

Ansatz 5 is $\Psi = \zeta(v)$ with inner variable $v = v_1 x + v_2 y + v_3 z + v_4 r$, $r^2 = x^2 + y^2 + z^2$, and second-order top element

$$L \;=\; A_2\,\partial^2 + L_1\,\partial + L_0, \qquad A_2 = a_0 + a_1 v,\; L_1 = b_0 + b_1 v,\; L_0 = c_0 + c_1 v,\; \partial = \tfrac{d}{dv}. \tag{4}$$

Write $v_{\mathrm{lin}} = (v_1, v_2, v_3)$ and $K = |v_{\mathrm{lin}}|^2 - v_4^2$. Since $v_{\mathrm{lin}} \cdot x = v - v_4 r$, the gradient and Laplacian of the inner variable are

$$|\nabla v|^2 = K + \frac{2v_4 v}{r}, \qquad \nabla^2 v = \frac{2v_4}{r},$$

so, for $\Psi = \zeta(v)$, $\nabla^2\Psi = \zeta''\big(K + \frac{2v_4 v}{r}\big) + \zeta'\,\frac{2v_4}{r}$. Clearing the Schrödinger equation $-\frac{1}{2}\nabla^2\Psi - \frac{1}{r}\Psi - E\Psi = 0$ by $-2r$ gives

$$\zeta''\,(Kr + 2v_4 v) + 2v_4\,\zeta' + 2\zeta + 2Er\,\zeta \;=\; 0. \tag{5}$$

Reducing ζ'' through L — multiply (5) by A_2 and substitute $A_2\zeta'' = -L_1\zeta' - L_0\zeta$ — and collecting on the *independent* radical r (valid where $|v_{\mathrm{lin}}| \neq 0$, so that v and r vary independently; Remark 2) leaves a remainder

$$r\,\big(R_1\zeta\big) + \big(R_0\zeta\big) \;=\; 0, \qquad h = A_2 = a_0 + a_1 v, \tag{6}$$

where the two first-order remainder operators are

$$R_1 = -KL_1\,\partial + \big(2EA_2 - KL_0\big), \tag{7}$$

$$R_0 = 2v_4\,(A_2 - vL_1)\,\partial + 2\,(A_2 - v_4 v L_0). \tag{8}$$

Because v and r are independent, (6) forces $R_1\zeta = 0$ and $R_0\zeta = 0$ separately. A partial stratum off the bad locus ($h \neq 0$) is thus a parameter point at which L, R_1, R_0 share a nonzero solution, i.e. $\mathrm{ord}\,\mathrm{GCRD}(L, R_1, R_0) = 1$.

Remark 2 (base-splitting genericity)**.** *The split on r requires v and r to be independent coordinates, i.e. $|v_{\mathrm{lin}}| \neq 0$; it also uses $A_2 \neq 0$ (the multiplier h) and, to keep R_1 of order one, $K \neq 0$ and $v_4 \neq 0$. The excluded loci $\{|v_{\mathrm{lin}}| = 0\}$ (radial inner variable, $v = v_4 r$), $\{K = 0\}$, $\{v_4 = 0\}$ and $\{h = 0\}$ are precisely the ODE-degeneration strata handled cell by cell by the differential Thomas decomposition; the radial locus $\{|v_{\mathrm{lin}}| = 0\}$ is where the hydrogen bound states live.*

3.2 The GCRD computation

With $\rho = -B_1/A_1$ the exponent of the R_1-mode ($A_1 = -KL_1,\ B_1 = 2EA_2 - KL_0$), a common solution of L, R_1, R_0 exists iff

(P) R_0 selects the same mode: $A_1B_0 - A_0B_1 = 0$ identically in v; and

(R) that mode solves L: $A_2(\rho' + \rho^2) + L_1\rho + L_0 = 0$ (the Riccati of L), cleared by A_1^2 to $A_2\big(B_1^2 - B_1'A_1 + B_1A_1'\big) - L_1B_1A_1 + L_0A_1^2 = 0$.

Each is an identity in v; equating v-coefficients gives polynomial equations in $(a_0, a_1, b_0, b_1, c_0, c_1, K, v_4, E)$. Let J be their ideal.

Theorem 1 (no genuine partial strata for ansatz 5 / hydrogen)**.** *Off the degenerate loci $\{K = 0\} \cup \{v_4 = 0\} \cup \{|v_{\text{lin}}| = 0\}$, the ansatz 5 system for the hydrogen Schrödinger equation has no partial-solution stratum with $h \not\equiv 0$. Consequently $V_\forall = V_{\exists\setminus\Psi}$ off the bad locus, and the entire gap $V_{\exists\setminus\Psi} \setminus V_\forall$ lies on the ODE-degeneration loci.*

Proof. Saturating J by K and v_4 (genericity) and then by (a_0, a_1) (imposing $h \not\equiv 0$) and computing a Gröbner basis, one finds the basis contains $b_0^2,\ b_0b_1,\ b_1^2$; hence every solution with $h \not\equiv 0$ lies on $\{b_0 = b_1 = 0\}$, i.e. $L_1 \equiv 0$. Saturating additionally by (b_0, b_1) — the branch $L_1 \neq 0$ — yields the unit ideal: there is no partial stratum with a genuine first-derivative term.

On the residual branch $L_1 \equiv 0$ the derivation above degenerates, since $A_1 = -KL_1 = 0$ and R_1 drops to order zero; it must be treated directly. There $R_1 = B_1$, so $R_1\zeta = 0$ forces $B_1 = 0$, i.e. $2EA_2 = KL_0$, whence $L_0 = (2E/K)A_2$ and, dividing L by $A_2 \neq 0$, the ODE is $\zeta'' + (2E/K)\zeta = 0$. The remaining operator is $R_0 = 2A_2\big(v_4\partial + 1 - (2Ev_4/K)v\big)$, giving $\rho_0 = \zeta'/\zeta = -1/v_4 + (2E/K)v$; requiring ρ_0 to solve the ODE's Riccati $\rho_0' + \rho_0^2 + 2E/K = 0$ and clearing by $(Kv_4)^2$ gives

$$4v_4^2E^2\,v^2 - 4Kv_4E\,v + 4Kv_4^2E + K^2 \;=\; 0 \quad (\text{in } v),$$

whose v^2-coefficient forces $E = 0$ and whose constant term then forces $K^2 = 0$, i.e. $K = 0$, excluded. So this branch is empty as well. Both branches being empty, no $h \not\equiv 0$ partial stratum exists; by (2) and (3), $V_\forall = V_{\exists\setminus\Psi}$ off the bad locus, and (1) leaves the gap on the degeneration loci. □

3.3 Reading

For ansatz 5 on hydrogen the second-order solution space never lets the Schrödinger equation peel off a single mode away from the degeneration loci: the potential's $-1/r$ term and the K, v_4 geometry conspire so that (P) and (R) are jointly unsatisfiable with $h \neq 0$. Consequently the membership computation, run *per Thomas cell*, already recovers the full homogeneous existence locus: the cells on the bad locus $\{h = 0\}$ and the radial locus $\{|v_{\text{lin}}| = 0\}$ — where the bound-state energies $E = -\frac{1}{2}, -\frac{1}{8}$ sit — are the only places $V_\forall$ and $V_{\exists\setminus\Psi}$ can differ, and those are exactly the degenerate cells the decomposition isolates. No separate GCRD $\forall$-cover (companion note II) is needed for ansatz 5; it remains the general safety net for ansätze whose GCRD does reach intermediate order.

Remark 3 (scope)**.** *Theorem 1 is specific to the* hydrogen *Hamiltonian and to ansatz 5's linear-coefficient second-order ODE. It does not follow from order-matching (companion note I gives a both-second-order counterexample with genuine strata) and does not transfer automatically: the helium Hamiltonian, and the quadratic-coefficient variants 5.1–5.3 and the coefficient-ring extensions 13–16, require re-running the same computation. The scripts* `ansatz5_gcrd.sage`/`ansatz5_gcrd2.sage` *are parameterized to make that a mechanical substitution.*

4 Summary

- $V_\forall = V_{\exists\setminus\Psi} \iff$ no partial strata $\iff$ (linear-ODE ansätze) $\operatorname{ord}\mathrm{GCRD}(L, \{R_\beta\}) \in \{0, n\}$ everywhere ((2), (3)).
- One-dimensional ansätze coincide unconditionally (Lemma 1); this covers the first-order and pure-exponential library families.
- Ansatz 5 on hydrogen has no genuine ($h \neq 0$) partial strata (Theorem 1); its $V_\forall$-vs-$V_{\exists\setminus\Psi}$ gap is entirely on the ODE-degeneration loci resolved by the Thomas decomposition, so per-cell membership recovers $V_{\exists\setminus\Psi}$ with no existential computation.
- Open: the log ansätze dimension check (2, 17, 17.1, 18, 19); the quadratic-coefficient and coefficient-ring second-order families; the nested (12) and product (10) ansätze; and the helium Hamiltonian.